



Day 7 Task - Operating System Forensics & Activity Reconstruction



Trios cyber • 10 Sept (Edited 12 Sept)

100/100 | Due 11 Sept, 15:00

Practical Assignment: Operating System Forensics & Activity Reconstruction

Dear Students,

For this practical assignment, you are required to examine operating-system artifacts and use available evidence to reconstruct basic user and system activity.

The objective of this exercise is to understand how operating-system artifacts can provide forensic information about users, applications, system activity and events.

Part 1: Operating System Identification

Identify the operating system associated with the provided system/image.

Select ONE of the following:

- Windows
- Linux
- macOS

Record:

- Operating system
- Version, where available
- Evidence/image source
- Relevant user/system information available in the evidence

Part 2: Examine Operating System Artifacts

Based on the operating system selected, examine relevant forensic artifacts.

For Windows, examine available artifacts such as:

- Windows Registry
- Event Logs
- User activity
- Program/application execution information



- Attached USB/device information

For Linux, examine available artifacts such as:

- System logs
- Command history
- User/system activity records

For macOS, examine available:

- Operating-system artifacts
- File-system information
- Application artifacts
- User activity records

If a particular artifact is not available, mention “Not Available.”

Part 3: Identify Evidence of Activity

Identify at least 3 artifacts that provide information about user or system activity.

For each artifact, record:

- Artifact name
- Location/path
- Type of information provided
- What you observed
- Why it may be relevant to an investigation

Take clear screenshots of the artifacts examined.

Part 4: Timeline & Activity Reconstruction

Use the available timestamps and artifacts to create a simple chronological sequence of events.

Your timeline should contain at least 5 events, where sufficient information is available.

For each event, record:

- Date/time
- Artifact/source
- Observed activity
- Interpretation

Clearly distinguish between:

- Direct evidence/observation
- Inference or interpretation

Part 5: Cross-Source Correlation

Compare at least two different artifact types.



For example:

- File metadata + event log
- File activity + application artifact
- User activity + command history
- Registry information + USB/device information

Explain whether the artifacts support or contradict each other.

Do not treat a single artifact as a complete explanation of an event.

Part 6: Document Your Work

Prepare a PDF or DOCX report containing:

1. Student Information
2. Objective
3. Operating system information
4. Artifacts examined
5. At least 3 relevant artifacts
6. Screenshots
7. Timeline of at least 5 events, where available
8. Cross-source correlation
9. Direct observations vs inferences
10. Conclusion

Each screenshot should have a short caption explaining what it demonstrates.

Submission Requirements

Submit ONE file only:

- PDF (.pdf) OR
- Microsoft Word document (.docx)

Important Notes

Use only an authorized forensic image or instructor-provided dataset.

Do not modify the forensic image or original evidence.

If an artifact is unavailable, clearly state "Not Available."

Your conclusion must be based on the available evidence. Do not state an inference as a confirmed fact.

Expected Learning Outcomes

After completing this practical, you should be able to:

1. Identify common operating-system forensic artifacts.
2. Examine user and system activity records.
3. Extract relevant information from OS artifacts.



4. Build a basic activity timeline.
5. Correlate evidence from multiple sources.
6. Distinguish direct observations from forensic inferences.
7. Present a defensible reconstruction of activity based on available evidence.

Submission: Upload your completed PDF/DOCX report to this Google Classroom assignment before the specified deadline. Late submissions are allowed only till after 48 hours from the time the task was posted but it may result in deduction of marks.

Please read the instructions carefully before beginning the assignment.

Your work

Marked

Day-7.pdf
PDF

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